

SEMESTER END / BACKLOG SUBJECT EXAMINATIONS - FEB / MARCH 2025

B.E. – CSE (Cyber Security) /			
Program	: CSE (Artificial Intelligence and Machine Learning)	Semester	: III
Course Name	: Computer Organization and Architecture	Max. Marks	: 100
Course Code	: CY34 / CI34 (2021,2022,2023 Batch)	Duration	: 3 Hrs.

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Using the basic theorems and postulates of Boolean algebra, simplify the following Boolean expression: $F = x'y'z + xyz + x'yz + xy'z$. CO1 (06)
 - Explain how a NAND gate can be used to synthesize any other basic logic gates. Demonstrate how to implement an AND, OR, and NOT gate using only NAND gates. CO1 (07)
 - What is a multiplexer? Design a 4-to-1 multiplexer and explain its working with a block diagram and truth table. CO1 (07)
- Design a 3-to-8 decoder and explain how it works with a block diagram and truth table. CO1 (08)
 - Design half subtractor and full subtractor. CO1 (06)
 - Explain the operation of an RS flip-flop, including its circuit diagram and truth table. CO1 (06)

UNIT - II

- Explain any Five data transfer instructions used by x86 architecture. CO2 (06)
 - List the difference between high-level language and machine language? CO2 (06)
 - Discuss Big endian with little endian with respect to byte addressability with an example. CO2 (08)
- With an example discuss the direct and indirect addressing modes. CO2 (06)
 - Explain assembly directive? List the few assembly directives and which directive are used to inform the assembler. CO2 (06)
 - With Example, explain the different ways in which the operands can be specified in an assembly language instruction. CO2 (08)

UNIT - III

- Illustrate the 4-bit carry-lookahead adder with a neat diagram. Explain how does it differs from a ripple-carry adder. CO3 (08)
 - Explain the difference between programmed I/O and interrupt-driven I/O. CO3 (07)
 - Compare the SCSI bus interface with the PCI bus in terms of functionality and use cases. CO3 (05)

6. a) Explain direct memory access (DMA), and how does it improve I/O performance? CO3 (06)
- b) Convert the following pairs of decimal numbers to 6 bit signed, 2's complement binary numbers and perform the addition and subtraction between them. State whether overflow occurs or not. CO3 (08)
- i) -14 and -9
 - ii) 6 and 11
 - iii) -13 and 18
 - iv) 15 and -10
- c) Describe how the SCSI bus facilitates communication with storage devices. CO3 (06)

UNIT- IV

7. a) Discuss the role of cache memory, main memory (RAM), and secondary storage in the memory hierarchy. CO4 (06)
- b) Compare the trade-offs between speed, cost, and capacity for each level of the hierarchy. CO4 (06)
- c) Given the following 4-bit binary numbers: $A=1101_2$ and $B=0110_2$. Multiply A and B using the standard binary multiplication method. Show the intermediate steps of the multiplication. CO4 (08)
8. a) Explain the working principles of FIFO (First-In-First-Out), LRU (Least Recently Used), and Random Replacement algorithms. Provide an example scenario where each algorithm might be most suitable. CO4 (08)
- b) Why is DRAM preferred for main memory, while SRAM is used for cache memory? CO4 (04)
- c) Perform the binary division of 110110_2 (dividend) by 101_2 (divisor) using the long division method. Show the step-by-step process of the division, including subtraction steps and how the quotient and remainder are determined. CO4 (08)

UNIT - V

9. a) What are the basic functions of a processing unit? Explain the role of the Control Unit in executing instructions. CO5 (07)
- b) Explain the difference between hard-wired control and microprogrammed control in the context of control units. CO5 (06)
- c) Describe the concept of multiple bus organization in a computer system. How does it improve the performance of instruction execution? CO5 (07)
- 10 a) Discuss the basic concepts of pipelining in processors. Explain the working of a RISC instruction pipeline and how it enhances performance. CO5 (08)
- b) Explain how to improve the performance in a pipelined system and discuss its limitations. CO5 (06)
- c) What are pipeline hazards? Explain the three types of hazards (structural, data, and control hazards). CO5 (06)
